Roll No. ....

Level 4, 1st Sem / DVOC (Ref. & Air Cond.)

Subject : Soldering & De-Soldering of Components-I

Time : 2 Hrs. M.M. : 50

SECTION-A

Note: Multiple Choice questions. All questions are compulsory. (5x1=5)

Q.1 What is the ideal temperature range for soldering electronic components?

a) 100-150° C b) 200-250°C

c) 300-350°C d) 500-600°C

Q.2 What is the main function of a de-soldering pump?

a) to apply solder to a joint

b) to remove excess solder from a joint

c) to clean the soldering iron tip

d) to heat up soldering components

Q.3 What type of solder is commonly used in electronics?

a) lead-free solder b) arc welding solder

c) cast iron solder d) Silver solder

Q.4 What does a cold solder joint look like?

a) Shiny and smooth b) dull and grainy

c) transparent d) Blackened and burnt

Q.5 Which method is commonly used to remove excess solder from a PCB?

a) Scratching with a blade
b) Using a de-soldering wick
c) Dipping in water
d) Pressing with a cloth

SECTION-B

Note: Objective type questions. All questions are compulsory. (5x1=5)

Q.6 What is the purpose of tinning a soldering iron tip?

Q.7 Name two common soldering defects.

Q.8 How can you identity a loose connection on a PCB?

Q.9 Why is it important to check PCB continuity after soldering.

Q.10 What are the advantages of using a temperature - controlled soldering station?

SECTION-C

Note: Short answer type questions. Attempt any six questions out of Eight questions. (6x5=30)

Q.11 What are the differences between single-layer and multi-layer PCBs?

Q.12 Explain the process of soldering SMD components.

Q.13 How does improper heat application affect a solder joint?

Q.14 What are the key safety precautions when working with a soldering iron?

Q.15 Describe the function of flux in soldering.

Q.16 What are the major factors that influence soldering quality?

Q.17 Explain the process of testing for broken tracks on a PCB.

Q.18 What are the advantages of lead-free solder?

SECTION-D

Note: Long answer type questions. Attempt any one questions out of two questions. (1x10=10)

Q.19 Discuss in detail the different soldering tools, their types. and their applications.

Q.20 Explain the process of soldering and de-soldering SMD components and the challenges involved.